

OMNI-GRAPH

Eradicating Unplanned Downtime

TEAM KONOHA

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The Knowledge Cliff

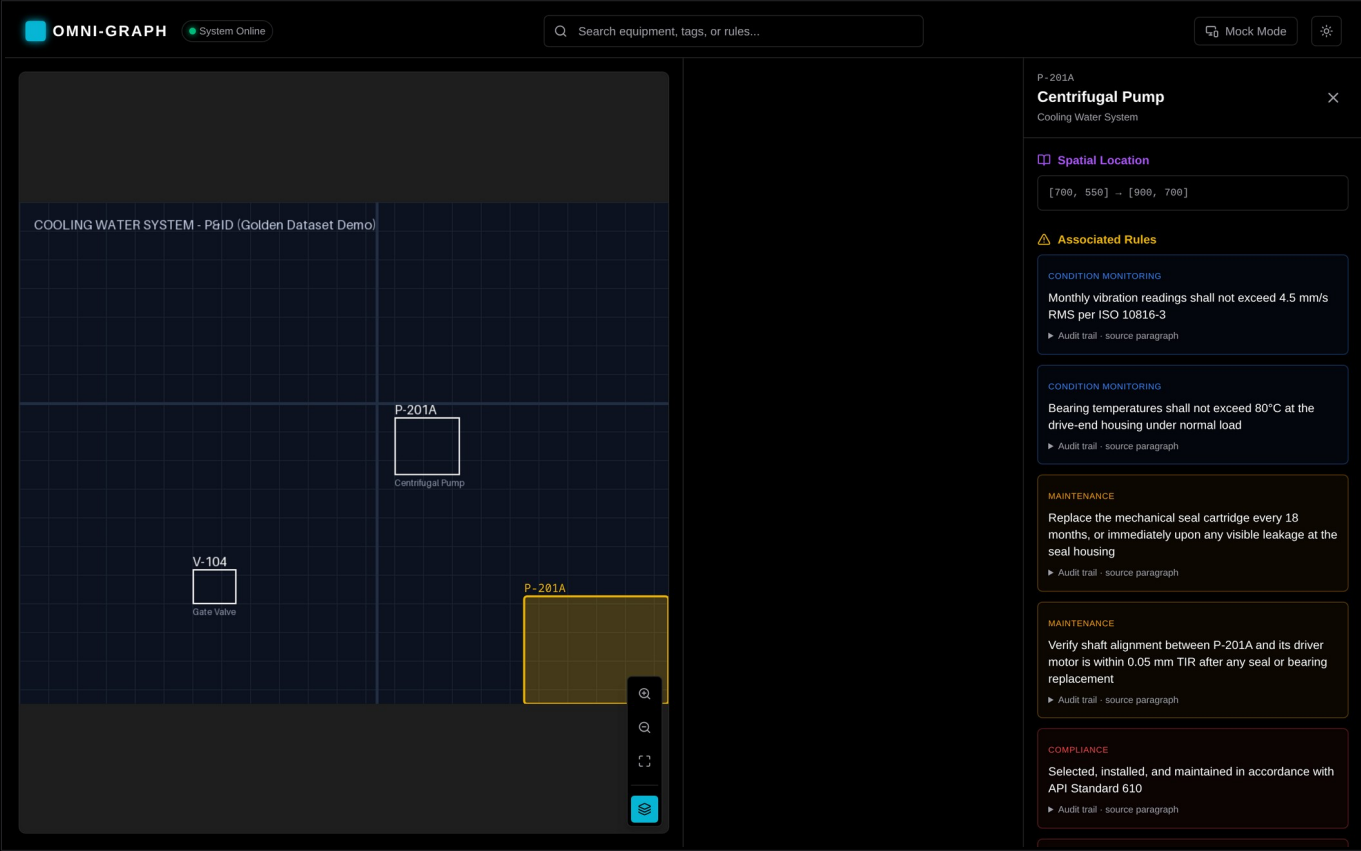
A technician facing an anomaly at 2 a.m. is often juggling **as many as a dozen disconnected systems** — PDFs, P&ID printouts, compliance binders, tribal memory — none of which point to each other.

Visakhapatnam, at night: a night-shift technician is flagged to pump P-201A. The senior engineer who wrote its maintenance SOP retired eighteen months ago. The manual is 400 pages, in a drawer, three buildings from the schematic on the wall.

"V-104"

**12 systems. 1
valve.
0 seconds to
find it.**

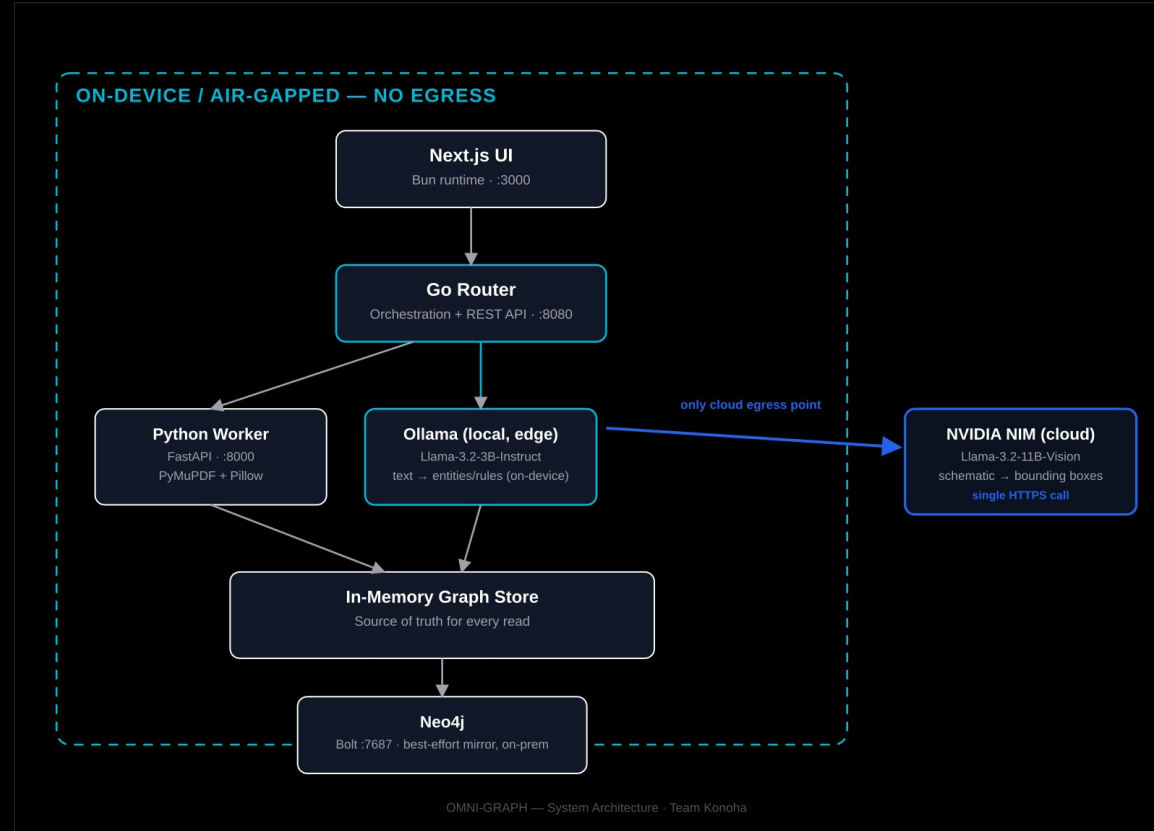
One Click. Two Realities, Instantly Synced.



- **2D Bounding-Box Canvas** — the real uploaded schematic, live overlays.
- **3D WebGL Knowledge Graph** — same assets, navigable force-directed graph.
- Click either pane — **both update in the same instant.**
- No vector search, no guessing — every answer traces to a real paragraph and a real coordinate.

activeNodeId: "P-201A"

Hybrid-Edge: Reason Locally. See in the Cloud.



- **Llama-3.2-3B + Neo4j run entirely on-device** — proprietary manuals never leave the plant network. Real air-gapped security, not a label.
- **NVIDIA NIM is the one deliberate cloud call** — only the task genuinely too heavy for a 4GB-VRAM edge box.
- **Deterministic Neo4j traversal beats vector RAG** — no embedding-similarity roulette.
- Vision link drops? **Falls back to a local replay path** instead of failing outright.

From Hours to Milliseconds

~~~40 min~~ → **<2 sec**

*"That's not an optimization. That's a category change."*

|                                  |                           |                    |
|----------------------------------|---------------------------|--------------------|
| KNOWLEDGE CLIFF                  | AUDIT TRAIL               | MARGINAL COST      |
| <b>Captured,<br/>Permanently</b> | <b>100%<br/>Traceable</b> | <b>~\$0 / seat</b> |